

# Elliot Paquette

Associate Professor

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## Employment & Professional affiliations

- 2023- **Associate professor (Tenured)**, *McGill University*, Mathematics & Statistics, Montreal, Quebec, Canada.
- 2025- **Associate member**, *MILA*, Montreal, Quebec, Canada.
- 2020-2023 **Assistant professor (Tenure stream)**, *McGill University*, Mathematics & Statistics, Montreal, Quebec, Canada.
- 2016-2020 **Assistant professor (Tenure stream)**, *The Ohio State University*, Mathematics & The Translational Data Analytics Institute, Columbus, Ohio, United States.
- 2013-2016 **NSF Postdoctoral Fellow**, *The Weizmann Institute of Science*, Faculty of Mathematics and Computer Science., Rehovot, Israel.

## Education

- 2008-2013 **Doctor of Philosophy**, *Mathematics*, University of Washington, Seattle, Washington, United States.  
Advisor: Ioana Dumitriu  
Thesis: *Eigenvalue fluctuations of random matrices beyond the Gaussian universality class*
- 2005-2008 **Bachelor of Arts, Magna cum Laude**, *Mathematics and Physics*, Kalamazoo College, Kalamazoo, Michigan, United States.  
Thesis: *A Synthetic Approach to the Characterization of Graph Invariants*

## Honours, grants, and funded visits

- Grant NSERC Discovery 2025-2030, RGPIN-2025-04643. (PI) \$355,000. *Scaling laws, high-dimensional limits of optimization algorithms, and random matrix theory.*
- Grant FRQNT-NOVA, 2025-2028. \$257,000. *Compute-Optimal Neural Scaling Laws for Learning Algorithms.* (co-PI, 33%)
- Grant Google Unrestricted Gift (October 2024). \$82,000. *Theory of compute optimality.*
- Grant CIFAR Catalyst grant 2025-2027. \$100,000. *Neural Scaling Laws and Compute-optimal Frontiers.* (co-PI, 33%)
- Visitor Long term participant, Institut Mittag Leffler. "Random matrices and scaling limits" Autumn 2024.
- Grant Google-Mila, 2023-2025: "Advancing scientific progress through AI fundamental research projects". (Collaborator) \$140,000. *High-dimensional Optimization for Machine Learning.* (Renewed once).
- Grant NSERC Discovery 2020-2025, RGPIN-2020-04974, and Discovery launch supplement. (PI) \$142,500. *Random series in the unit disk, random matrix theory, and the Gaussian multiplicative chaos.*
- Grant McGill University startup fund. (PI) \$100,000.
- Grant USA-Israel Binational Science Foundation Startup Grant, no. 2018341, with Alon Nishry of Tel Aviv University (2019). US\$60,000. *Gaussian analytic functions and planar branching processes.* 2019-2021 (ended early on account of relocation)
- Grant Simons travel grant No. 638152. US\$40,000. 2019-2024 (ended early on account of relocation)

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## Instruction

- Fall 2026 Math 598/784. Topics in Probability: High dimensional probability. 25 students.
- Summer 2026 Instructor at the “Princeton Machine Learning Theory Summer School” in Princeton, New Jersey. August 2026.
- Winter 2025 Math 547. Honours stochastic processes. 27 students.
- Winter 2025 Math 447. Introduction to stochastic processes. 107 students.
- Summer 2024 Instructor at the “CRM-PIMS summer school in probability” in Montreal. July 1-26 2024.
- Summer 2024 Instructor at the “Random matrix theory summer school” at the University of Michigan. June 17-28 2024.
- Winter 2024 Reading course (Math 698): algorithmic spin glasses. 3 students registered. 15 participants.
- Winter 2024 Math 589. Advanced Probability II. 11 students.
- Fall 2023 Math 598. Topics in Probability: High dimensional probability. 20 students.
- Summer 2023 Summer school speaker in probability and optimization. Lehigh University. Available [here](#).
- Winter 2023 Math 547. Stochastic processes. 14 students. Developed new course notes. Available [here](#).
- Fall 2022 Math 356. Honours Probability. 35 students.
- Fall 2022 Coordinator for Math 222. Calculus 3. Total 400 students. In my section 200 students.
- Winter 2022 Math 447, Stochastic Processes. 130 students.
- Winter 2022 Math 598/784, Topics in Probability and Statistics (Random matrix theory). 20 students.
- Fall 2021 Math 356, Honours Probability. Measure theoretic probability for undergraduates. 70 students.
- Winter 2021 Math 447, Stochastic Processes. Course on Markov chains and some continuous time processes. Predominantly U3, in-major, 170 students.
- Fall 2020 Math 240, Discrete Mathematics. Developed webwork assignments and new tutorials. Majority CS/CSE, mostly U2. 250 students.
- 2016-2019 At the Ohio State University: an instructor for Math 3356 introductory proof course, Math 6251-2 graduate probability sequence, Math 2255 ordinary differential equations.

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## Organization

- Conference Workshop organizer for “High-dimensional learning dynamics,” Lead organizer 2026. (Co-organizer 2023). ICML. This contained a non-archival proceedings, with submissions reviewed and 6 merit talks awarded. 5 plenary speakers. 100+ participants.
- Conference Summer school co-organizer for “Universal statistics in number theory,” May 2026. CRM.
- Conference BIRS Program “Large- $N$  Matrix Models and Emergent Geometry,” May 2026. CRM.
- Conference Workshop organizer for “Phase transitions and dynamics in random media,” June 2025. CRM.
- Conference Workshop organizer for “Random matrix theory and high-dimensional learning dynamics,” June 2025. CRM thematic program in “Mathematical foundations of data science.”
- Conference Workshop organizer for “The many facets of random matrix theory,” December 2023. Winter CMS Meeting 2023. 20 participants.
- Seminar (2021+) Organizer for the RMT-OPT-ML seminar (random matrix theory, optimization, and machine learning). Weekly seminar for presentation of recent advances in theory of optimization, and random matrix theory. 20 regular participants.
- Seminar (2021+) Organizer for the CRM-ISM probability seminar. 20 regular participants.

- Summer program (2021) Organizer and instructor (with Courtney Paquette) for the RMT-OPT (random matrix theory and optimization) summer program. 7 students (5 BSc and 2 postgraduate) participated in the summer program, split into 3 groups. Financially supported by 1 NSERC USRA award, 1 McGill SURA award, 1 ISM undergraduate summer scholar, and NSERC Discovery grant (listed above). Began as a 4 week minicourse, and concluded with 3 groups working on research around random matrix theory and optimization.
- Conference Organizer for “Algebraic questions in random integral matrices” conference, March 28-29 2020. Rescheduled for November 14-15, 2020 as an online conference. 50 participants.
- Conference Organizer for “Stochastic Spatial Processes at OSU” conference, April 18-19 2020. Rescheduled for March 2021 as an online conference. 50 participants.

## Committees & Scholarship

- Committee NSERC RTI program 2026: Computer, Mathematical and Statistical Sciences Selection committee panelist.
- Conference ref. ICML 2025-2026 Area chair. ICML Workshop reviewer 2024-2025.
- Conference ref. ICLR 2025 reviewer.
- Conference ref. NeurIPS top reviewer 2023. Area chair 2024-2025.
- Committee NSERC Faculty level CGS-M Evaluation committee, 2023, 2024, & 2025
- Committee NSERC CGS-M Departmental evaluation committee chair, 2023 & 2024
- Committee (2021) CRM-ISM postdoctoral selection committee.
- Committee (2021-24) McGill Department Committee for Graduate Affairs. Prize subcommittee chair.
- Reviewer (2022-) External reviewer for grant organizations: NSERC Discovery, NWO (Dutch research council), ISF (Israel Science Foundation), FWO (Flemish research council).
- Thesis examiner (2024) University of Toronto, (2022+) McGill Mathematics, (2021) Weizmann Institute Mathematics
- Referee Journal of the AMS. Inventiones Mathematicae. Duke Journal of Mathematics. International Mathematics Research Notices. Transactions of the AMS. Communications in Mathematical Physics. Communications in Pure and Applied Mathematics. Proceedings of the London Mathematical Society. Israel Journal of Mathematics. Annals of Probability. Annals of Applied Probability. Electronic Journal of Probability. Random Structures & Algorithms. Discrete & Computational Geometry.

## Current Students & Postdocs

- Postdoc Deborah Sampaio. (2025-). McGill University.
- PhD Kevin Xiao. (2023-). McGill University.
- PhD Noah Marshall. (2023-). McGill University.
- PhD Maria Esipova. *Joint with Louigi Addario-Berry.*(2025-). McGill University.
- MSc Yixi Aurora Wang. *Joint with Courtney Paquette.* (2024-). McGill University.
- MSc Brayden Smith. (2025-). McGill University.

## Students & Postdocs mentored

- Postdoc Elizabeth Collins-Woodfin. (2022-2025) Postdoctoral fellow. McGill & Mila. currently: Assistant Professor University of Oregon.
- Postdoc Inbar Seroussi. (2024-2024) Postdoctoral fellow. McGill & Mila. currently: Assistant Professor Tel Aviv University.
- Postdoc Érika Roldán. (2019–2020) Postdoctoral fellow. The Ohio State University. currently: Max Planck Fellow.
- PhD Vincent Painchaud. (2021–2026). McGill University. currently: Postdoc KTH Royal Institute of Technology (Stockholm)

- PhD Kiwon Lee. (2020–2023). McGill University. currently: Lecturer at the Computational Data Science Institute at McGill University
- PhD Andrew Vander Werf. (2018–2023). The Ohio State University. currently: Postdoc Brown University
- MSc Hugo Latourelle–Vigeant. *Joint with Courtney Paquette*. (2022–2024). McGill University. currently: PhD Yale University
- MSc Samuel Kirkiles. (2023–2024). McGill University. currently: Software engineer, Coinbase
- MSc Andrew Cheng. *Joint with Courtney Paquette*. (2022–2023). McGill University. currently: PhD Harvard University
- MSc Romain Pagès. (2021–2022). KTH (Stockholm). Research assistant at McGill. Currently: Stagiaire de recherche, INRAE, Paris.
- MSc Kiwon Lee. (2018–2020). The Ohio State University. Currently: PhD student McGill (listed above).
- BSc R.E. Austin Huang (SURA). 2023.
- BSc R.E. Kevin Zhao (ARIA). 2023.
- BSc R.E. Samuel Kirkles (SURA). 2022.
- BSc R.E. Hugo Latourelle-Vigeant (USRA). 2021.
- BSc R.E. Nicolas Fertout. 2021. Subsequently: MSc Stanford
- BSc R.E. Vincent Savignac. 2021. Subsequently: MSc McGill
- BSc R.E. Jiajun Yu (SURA). 2021. Subsequently: MSc Oxford
- BSc R.E. Ria Stevens (SURA). 2021. Subsequently: PhD Rice.
- BSc Thesis Anna Brandenberger. 2020. Subsequently: PhD MIT.

## Publications and Conference Proceedings

### Preprints

- 2026 61. Katie Everett and Elliot Paquette. “Dynamics of Stochastic Momentum with Sparse Updates in High Dimensions”. In: *arXiv e-prints* (May 2026). arXiv: [2605.28961](#) [[stat.ML](#)] DOI
60. Elliot Paquette, Noah Marshall, Lucas Benigni, Guangyuan Wang, Atish Agarwala, and Courtney Paquette. “Phases of Muon: When Muon Eclipses SignSGD”. in: *arXiv e-prints* (May 2026). arXiv: [2605.09552](#) [[math.OC](#)] DOI
59. Andy Zeyi Liu, Elliot Paquette, and John Sous. “Spectral Lens: Activation and Gradient Spectra as Diagnostics of LLM Optimization”. In: *arXiv e-prints* (May 2026). arXiv: [2605.05683](#) [[stat.ML](#)] DOI
58. Elliot Paquette, Ke Liang Xiao, and Yizhe Zhu. “Power-Law Spectrum of the Random Feature Model”. In: *arXiv e-prints* (Mar. 2026). arXiv: [2603.14578](#) [[stat.ML](#)] DOI
57. Zhou Fan, Renyuan Ma, Elliot Paquette, and Zhichao Wang. “Anisotropic local law for non-separable sample covariance matrices”. In: *arXiv e-prints* (Feb. 2026). arXiv: [2602.17960](#) [[math.PR](#)]
56. Damien Ferbach, Courtney Paquette, Gauthier Gidel, Katie Everett, and Elliot Paquette. “Logarithmic-time Schedules for Scaling Language Models with Momentum”. In: *arXiv e-prints* (Feb. 2026). arXiv: [2602.05298](#) [[stat.ML](#)] DOI
- 2025 55. Shurong Lin, Eric D. Kolaczyk, Adam Smith, and Elliot Paquette. “High-Dimensional Privacy-Utility Dynamics of Noisy Stochastic Gradient Descent on Least Squares”. In: *arXiv e-prints* (Oct. 2025). arXiv: [2510.16687](#) [[stat.ML](#)] DOI

54. Lucas Benigni and Elliot Paquette. “Eigenvalue distribution of the Neural Tangent Kernel in the quadratic scaling”. In: *arXiv e-prints* (Aug. 2025). arXiv: [2508.20036](#) [\[math.PR\]](#) DOI
53. Gaultier Lambert and Elliot Paquette. “Bulk asymptotics of the Gaussian  $\beta$ -ensemble characteristic polynomial”. In: *arXiv e-prints* (Aug. 2025). arXiv: [2508.01458](#) [\[math.PR\]](#) DOI
52. Joseph Najnudel, Elliot Paquette, Nick Simm, and Truong Vu. “The Fourier coefficients of the holomorphic multiplicative chaos in the limit of large frequency”. In: *arXiv e-prints* (Feb. 2025). arXiv: [2502.14863](#) [\[math.PR\]](#) DOI
51. Vincent Painchaud and Elliot Paquette. “Operator level soft edge to bulk transition in  $\beta$ -ensembles via canonical systems”. In: *arXiv e-prints* (Feb. 2025). arXiv: [2502.10305](#) [\[math.PR\]](#) DOI
- 2024 50. Alon Nishry and Elliot Paquette. “The image of random analytic functions: coverage of the complex plane via branching processes”. In: (2024). arXiv: [2403.19380](#) [\[math.PR\]](#)
- 2020 49. Gaultier Lambert and Elliot Paquette. “Strong approximation of Gaussian  $\beta$ -ensemble characteristic polynomials: the edge regime and the stochastic Airy function”. In: *arXiv e-prints* (Sept. 2020). arXiv: [2009.05003](#) [\[math.PR\]](#)

#### Refereed Conference Proceedings

- 2025 48. Damien Ferbach, Katie Everett, Gauthier Gidel, Elliot Paquette, and Courtney Paquette. “Dimension-adapted Momentum Outpaces SGD”. in: *NeurIPS Spotlight* (Dec. 2025). arXiv: [2505.16098](#) [\[stat.ML\]](#) DOI
47. Ke Liang Xiao, Noah Marshall, Atish Agarwala, and Elliot Paquette. “Exact Risk Curves of signSGD in High-Dimensions: Quantifying Preconditioning and Noise-Compression Effects”. In: *ICML* (2025). arXiv: [2411.12135](#) [\[stat.ML\]](#) DOI
46. Noah Marshall, Ke Liang Xiao, Atish Agarwala, and Elliot Paquette. “A Clipped Trip: the Dynamics of SGD with Gradient Clipping in High-Dimensions”. In: *ICLR* (2025). arXiv: [2406.11733](#) [\[stat.ML\]](#)
- 2024 45. Elizabeth Collins-Woodfin, Inbar Seroussi, Begoña García Malaxechebarría, Andrew W. Mackenzie, Elliot Paquette, and Courtney Paquette. “The High Line: Exact Risk and Learning Rate Curves of Stochastic Adaptive Learning Rate Algorithms”. In: *NeurIPS* (2024). arXiv: [2405.19585](#) [\[math.OC\]](#)
44. Elliot Paquette, Courtney Paquette, Lechao Xiao, and Jeffrey Pennington. “4+3 Phases of Compute-Optimal Neural Scaling Laws”. In: *NeurIPS Spotlight* (2024). arXiv: [2405.15074](#) [\[stat.ML\]](#)
- 2023 43. Kiwon Lee, Andrew N. Cheng, Courtney Paquette, and Elliot Paquette. “Trajectory of Mini-Batch Momentum: Batch Size Saturation and Convergence in High Dimensions”. In: *Advances in Neural Information Processing Systems*. Vol. 35. June 2022. arXiv: [2206.01029](#) [\[math.OC\]](#)
42. Courtney Paquette, Elliot Paquette, Ben Adlam, and Jeffrey Pennington. “Implicit Regularization or Implicit Conditioning? Exact Risk Trajectories of SGD in High Dimensions”. In: *Advances in Neural Information Processing Systems*. Vol. 35. June 2022. arXiv: [2206.07252](#) [\[stat.ML\]](#)

- 2021 41. Courtney Paquette and Elliot Paquette. “Dynamics of Stochastic Momentum Methods on Large-scale, Quadratic Models”. In: *Advances in Neural Information Processing Systems*. Vol. 34. Curran Associates, Inc., 2021. arXiv: [2106.03696](#) [math.OC]
40. Courtney Paquette, Kiwon Lee, Fabian Pedregosa, and Elliot Paquette. “SGD in the Large: Average-case Analysis, Asymptotics, and Stepsize Criticality”. In: *Proceedings of Thirty Fourth Conference on Learning Theory*. Vol. 134. Proceedings of Machine Learning Research. PMLR, Aug. 2021. arXiv: [2102.04396](#) [math.OC]

### Print publications

- 2026 39. Hugo Latourelle-Vigeant and Elliot Paquette. “Dyson equation for correlated linearizations and test error of random features regression”. In: *Random Matrices Theory Appl.* 15.01 (2026). arXiv: [2312.09194](#) [math.ST] DOI
- 2025 38. Courtney Paquette, Elliot Paquette, Ben Adlam, and Jeffrey Pennington. “Homogenization of SGD in high-dimensions: exact dynamics and generalization properties”. In: *Math. Program.* 214.1-2 (2025). arXiv: [2205.07069](#) [math.ST] DOI
37. Elliot Paquette and Ofer Zeitouni. “The extremal landscape for the  $C\beta E$  ensemble”. In: *Forum Math. Sigma* 13 (2025). arXiv: [2209.06743](#) [math.PR] DOI
- 2024 36. Elizabeth Collins-Woodfin, Courtney Paquette, Elliot Paquette, and Inbar Seroussi. “Hitting the High-dimensional notes: an ODE for SGD learning dynamics on GLMs and multi-index models”. In: *Information and Inference: A Journal of the IMA* 13.4 (Oct. 2024). arXiv: [2308.08977](#) [math.OC] DOI
35. Afonso S. Bandeira, Antoine Maillard, Shahar Mendelson, and Elliot Paquette. “Fitting an ellipsoid to a quadratic number of random points”. In: *ALEA* 21 (2024). arXiv: [2307.01181](#) [math.PR] DOI
34. Shurong Lin, Elliot Paquette, and Eric D. Kolaczyk. “Differentially Private Linear Regression with Linked Data”. In: *Harvard Data Science Review* 6.3 (July 2024). arXiv: [2308.00836](#) [stat.ME] DOI
33. Elizabeth Collins–Woodfin and Elliot Paquette. “High-dimensional limit of one-pass SGD on least squares”. In: *Electronic Communications in Probability* 29 (2024). arXiv: [2304.06847](#) [math.PR] DOI
- 2023 32. Joseph Najnudel, Elliot Paquette, and Nick Simm. “Secular coefficients and the holomorphic multiplicative chaos”. In: *Ann. Probab.* 51.4 (2023). arXiv: [2011.01823](#) [math.PR] DOI
31. Andrew Newman and Elliot Paquette. “The integer homology threshold in  $Y_d(n, p)$ ”. In: *Proc. Amer. Math. Soc.* 151.8 (2023). arXiv: [1808.10647](#) [math.CO] DOI
30. Tim Hoheisel and Elliot Paquette. “Uniqueness in nuclear norm minimization: flatness of the nuclear norm sphere and simultaneous polarization”. In: *J. Optim. Theory Appl.* 197.1 (2023). arXiv: [2205.08442](#) [math.CV] DOI
29. Alon Nishry and Elliot Paquette. “Gaussian analytic functions of bounded mean oscillation”. In: *Analysis & PDE* 16.1 (Apr. 2023). arXiv: [2002.00804](#) [math.CV] DOI
28. Gaultier Lambert and Elliot Paquette. “Strong approximation of Gaussian beta-ensemble characteristic polynomials: the hyperbolic regime”. In: *Ann. Appl. Probab.* 33.1 (2023). arXiv: [2001.09042](#) [math.PR] DOI



- 2022 27. Elliot Paquette and Thomas Trogdon. "Universality for the conjugate gradient and MINRES algorithms on sample covariance matrices". In: *Communications in Pure and Applied Mathematics* (Sept. 2022). arXiv: [2007.00640](#) [[math.NA](#)]
26. Pascal Maillard and Elliot Paquette. "Interval fragmentations with choice: Equidistribution and the evolution of tagged fragments". In: *The Annals of Applied Probability* 32.5 (2022). arXiv: [2006.16932](#) [[math.PR](#)] DOI
25. Courtney Paquette, Bart van Merriënboer, Elliot Paquette, and Fabian Pedregosa. "Halting Time is Predictable for Large Models: A Universality Property and Average-case Analysis". In: *Foundations of Computational Mathematics* (June 2022). arXiv: [2006.04299](#) [[math.OC](#)] DOI
- 2021 24. Itai Benjamini, Yoav Krauz, and Elliot Paquette. "Anchored expansion of Delaunay complexes in real hyperbolic space and stationary point processes". In: *Probability Theory and Related Fields* (Nov. 2021). arXiv: [2008.01063](#) [[math.PR](#)] DOI
23. Matthew Kahle, Elliot Paquette, and Érika Roldán. "The fundamental group of 2-dimensional random cubical complexes". In: *Forum of Mathematics, Sigma* 9 (2021). arXiv: [2001.07812](#) [[math.CO](#)] DOI
22. Matías Carrasco, Pablo Lessa, and Elliot Paquette. "On the speed of distance stationary sequences". In: *ALEA* 18 (2021). arXiv: [1912.12523](#) [[math.PR](#)]
- 2020 21. Anirban Basak, Elliot Paquette, and Ofer Zeitouni. "Spectrum of random perturbations of Toeplitz matrices with finite symbols". In: *Trans. Amer. Math. Soc.* 373.7 (2020). arXiv: [1812.06207](#) [[math.PR](#)] DOI
20. Hoi H. Nguyen and Elliot Paquette. "Surjectivity of near-square random matrices". In: *Combinatorics, Probability and Computing* 29.2 (2020). arXiv: [1802.00001](#) [[math.ST](#)] DOI
- 2019 19. Anirban Basak, Elliot Paquette, and Ofer Zeitouni. "Regularization of non-normal matrices by Gaussian noise - the banded Toeplitz and twisted Toeplitz cases". In: *Forum Math. Sigma* 7 (Nov. 2017). eprint: [1712.00042](#) ([math.PR](#)) DOI
18. Christopher Hoffman, Matthew Kahle, and Elliot Paquette. "Spectral Gaps of Random Graphs and Applications". In: *International Mathematics Research Notices* (May 2019). eprint: [1201.0425](#) DOI
- 2018 17. Elliot Paquette. "Distributional Lattices on Riemannian symmetric spaces". In: *Unimodularity in randomly generated graphs*. Vol. 719. Contemp. Math. Amer. Math. Soc., Providence, RI, 2018. arXiv: [1707.00308](#) [[math.PR](#)] DOI
16. Diane Holcomb and Elliot Paquette. "The maximum deviation of the  $Sine_\beta$  counting process". In: *Electron. Commun. Probab.* 23 (2018). eprint: [1801.08989](#) DOI
15. Itai Benjamini, Elliot Paquette, and Joshua Pfeffer. "Anchored expansion, speed and the Poisson–Voronoi tessellation in symmetric spaces". In: *Annals of Probability* 46.4 (July 2018). arXiv: [1409.4312](#) [[math.PR](#)] DOI
14. Gaultier Lambert and Elliot Paquette. "The law of large numbers for the maximum of almost Gaussian log-correlated fields coming from random matrices". In: *Probability Theory and Related Fields* (Feb. 2018). arXiv: [1611.08885](#) [[math.PR](#)] DOI
13. Ioana Dumitriu and Elliot Paquette. "Spectra of Overlapping Wishart Matrices and the Gaussian Free Field". In: *Random Matrix Theory and Applications* 7.02 (2018). arXiv: [1410.7268](#) [[math.PR](#)] DOI

12. Elliot Paquette and Ofer Zeitouni. "The maximum of the CUE field". In: *International Mathematics Research Notices* 2018.16 (2018). arXiv: [1602.08875 \[math.PR\]](#) DOI
- 2017 11. Elliot Paquette and Ofer Zeitouni. "Extremal eigenvalue correlations in the GUE minor process and a law of fractional logarithm". In: *The Annals of Probability* 45.6A (2017). arXiv: [1505.05627 \[math.PR\]](#) DOI
10. Elliot Paquette and Younghwan Son. "Birkhoff sum fluctuations in substitution dynamical systems". In: *Ergodic Theory and Dynamical Systems* (2017). arXiv: [1505.01428 \[math.DS\]](#) DOI
9. Christopher Hoffman, Matthew Kahle, and Elliot Paquette. "The threshold for integer homology in random d-complexes". In: *Discrete & Computational Geometry* 57.4 (2017). arXiv: [1308.6232 \[math.AT\]](#) DOI
- 2016 8. Pascal Maillard and Elliot Paquette. "Choices and intervals". In: *Israel J. Math.* 212.1 (2016). eprint: [1402.3931](#) DOI
- 2015 7. Ohad Noy Feldheim, Elliot Paquette, and Ofer Zeitouni. "Regularization of non-normal matrices by Gaussian noise". In: *Int. Math. Res. Not. IMRN* 18 (2015). eprint: [1404.3491](#) DOI
6. Yury Malyskin and Elliot Paquette. "The power of choice over preferential attachment". In: *ALEA Lat. Am. J. Probab. Math. Stat.* 12.2 (2015). eprint: [1311.1091](#)
- 2014 5. Yury Malyskin and Elliot Paquette. "The power of choice combined with preferential attachment". In: *Electron. Commun. Probab.* 19 (2014). eprint: [1403.4301](#) DOI
- 2013 4. Ioana Dumitriu, Tobias Johnson, Soumik Pal, and Elliot Paquette. "Functional limit theorems for random regular graphs". In: *Probab. Theory Related Fields* 156.3-4 (2013). eprint: [1109.4094](#) DOI
- 2012 3. Ioana Dumitriu and Elliot Paquette. "Global fluctuations for linear statistics of  $\beta$ -Jacobi ensembles". In: *Random Matrices Theory Appl.* 1.4 (2012). eprint: [1203.6103](#) DOI
- 2010 2. Tamás Keleti and Elliot Paquette. "The trouble with von Koch curves built from  $n$ -gons". In: *Amer. Math. Monthly* 117.2 (2010) DOI
- 2009 1. Elliot Paquette and Christopher Seaton. "The index of a vector field on an orbifold with boundary". In: *Involve* 2.2 (2009). eprint: [0806.2113](#) DOI

### Other articles

- Courtney Paquette and Elliot Paquette. "High-dimensional optimization". In: *SIAM Views and News* 20 (Dec. 2022)
- Elliot Paquette and Andrew Vander Werf. "Random geometric graphs and the spherical Wishart matrix". In: *arXiv e-prints* (Oct. 2021). arXiv: [2110.10785 \[math.PR\]](#)
- Matías Carrasco, Pablo Lessa, and Elliot Paquette. "A Furstenberg type formula for the speed of distance stationary sequences". In: *submitted* (Oct. 2017). arXiv: [1710.00733 \[math.PR\]](#)
- Diane Holcomb and Elliot Paquette. "Tridiagonal Models for Dyson Brownian Motion". In: *submitted* (July 2017). arXiv: [1707.02700 \[math.PR\]](#)



○ Tobias Johnson and Elliot Paquette. “Quantitative Small Subgraph Conditioning”. In: *ArXiv e-prints* (July 2013). arXiv: [1307.4858 \[math.PR\]](#)

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## Invited Talks Given (2017-present)

- March 2026 ○ E. Paquette. *Local laws for sample covariance matrices beyond the separable case*. Princeton Probability seminar, Princeton, New Jersey.
- February 2026 ○ E. Paquette. *The Circular Beta Ensemble: Magic, Mystery, and the Extremes of Random Matrices*. **CRM-ISM Seminar**, Montréal.
- January 2026 ○ E. Paquette. *Local laws for sample covariance matrices beyond the separable case*. Weizmann Probability seminar, Rehovot, Israel.
- November 2025 ○ E. Paquette. *Power law data covariance and a solvable model of Kaplan scaling laws*. McMaster Probability seminar, Hamilton, Ontario.
- October 2025 ○ E. Paquette. *Power law data covariance and a solvable model of Kaplan scaling laws*. Georgia Tech Probability seminar, Atlanta.
- October 2025 ○ E. Paquette. *From magic squares, through random matrices, and to the multiplicative chaos*. University of Wisconsin Probability seminar, Madison, Wisconsin.
- October 2025 ○ E. Paquette. *From magic squares, through random matrices, and to the multiplicative chaos*. UPenn-Temple Probability seminar, Philadelphia.
- September 2025 ○ E. Paquette. *Power law data covariance and a solvable model of Kaplan scaling laws*. **AI+Physics Conference**, The CUNY Graduate Center, New York.
- August 2025 ○ E. Paquette. *Power law data covariance and a solvable model of Kaplan scaling laws*. **Statistical physics and machine learning: moving forward**, Cargèse, France.
- June 2025 ○ E. Paquette. *High-dimensional Optimization with Applications to Compute-Optimal Neural Scaling Laws*. CMS summer special session: Mathematics of Machine Learning, Université de Laval, Quebec (*Cancelled due to injury*).
- June 2025 ○ E. Paquette. *From magic squares, through random matrices, and to the multiplicative chaos*. CMS summer special session: Analysis and probability, and their interactions, Université de Laval, Quebec (*Cancelled due to injury*).
- Jun 2025 ○ E. Paquette. *From magic squares, through random matrices, and to the multiplicative chaos*. **RMTA Online seminar**, Chinese online random matrix theory association.
- April 2025 ○ E. Paquette. *High-dimensional Optimization with Applications to Compute-Optimal Neural Scaling Laws*. **Yale Foundations of Data Science Colloquium**, Yale University.
- February 2025 ○ E. Paquette. *From magic squares, through random matrices, and to the multiplicative chaos*. **Montréal-Guanajuato Workshop in Probability and machine learning**, CIMAT, Mexico.
- December 2024 ○ E. Paquette. *Random matrix theory for high dimensional optimization, and an application to scaling laws*. Probability and statistics seminar. USC.
- December 2024 ○ E. Paquette. *Kahane’s coverage question and the image of Gaussian analytic function*. Probability seminar. UC San Diego.
- October 2024 ○ E. Paquette. *Kahane’s coverage question and the image of Gaussian analytic function*. **Workshop on random matrices and scaling limits**, Institut Mittag Leffler, Stockholm.

- May 2024 [○](#) E. Paquette. *The random matrix Fyodorov-Hiary-Keating conjecture*. **Random matrices and related topics in Jeju**, Jeju Island, Korea.
- April 2024 [○](#) E. Paquette. *Random matrix theory for high dimensional optimization, and an application to scaling laws*. **Vector Distinguished Lecture Series**, Vector Institute, Toronto.
- January 2024 [○](#) E. Paquette. *High dimensional limits of SGD*. **Mathematics of Data Science Conference**, NUS Singapore.
- June 2023 [○](#) E. Paquette. *The extremal landscape of the circular  $\beta$ -ensemble*. Probability seminar. Université de Lille.
- June 2023 [○](#) E. Paquette. *The extremal landscape of the circular  $\beta$ -ensemble*. Probability seminar. University of Sussex.
- May 2023 [○](#) E. Paquette. *The scaling limits of the Gaussian beta ensemble characteristic polynomial*. **Midrasha on random Schrödinger operators and random matrices**. Israel Institute Advanced Study, Jerusalem, Israel.
- April 2023 [○](#) E. Paquette. *The homogenization of SGD in high dimensions*. Probability seminar. KTH, Stockholm, Sweden.
- April 2023 [○](#) E. Paquette. *The homogenization of SGD in high dimensions*. DACO seminar. ETH, Zürich.
- March 2023 [○](#) E. Paquette. *The extremal landscape of the circular  $\beta$ -ensemble*. Probability seminar. Cornell University.
- March 2023 [○](#) E. Paquette. *The homogenization of SGD in high dimensions*. Probability seminar. University of Waterloo, Ontario.
- January 2023 [○](#) E. Paquette. *Homogenization of SGD in the Large*. **Topics in High Dimensional Probability**. ICTS-TIFR, Bangalore, India.
- November 2022 [○](#) E. Paquette. *The extremal landscape of the circular  $\beta$ -ensemble*. **Colloquium** University of Toronto.
- November 2022 [○](#) E. Paquette. *Homogenization of SGD in the Large*. Lehigh–Minnesota Probability seminar Lehigh & University of Minnesota.
- October 2022 [○](#) E. Paquette. *Homogenization of SGD in the Large*. Lefschetz dynamical systems seminar Brown University.
- October 2022 [○](#) E. Paquette. *The extremal landscape of the circular  $\beta$ -ensemble*. Large deviations and random media Conference. NYU.
- June 2022 [○](#) E. Paquette. *Optimization Algorithms in the Large: Generalization and Universality*. Conference on Random Matrix Theory and Numerical Linear Algebra. Seattle.
- Dynamics 2021 [○](#) E. Paquette. *Dynamics of Stochastic Momentum Methods on Large-scale, Quadratic Models*. NeurIPS '21. Vancouver.
- November 2021 [○](#) E. Paquette. *The edge scaling limit of the Gaussian beta-ensemble characteristic polynomial*. Probability seminar. University of Washington.
- October 2021 [○](#) E. Paquette. *The threshold for simple-connectedness in hypercube percolation*. Probability seminar. University of Groningen.
- August 2021 [○](#) E. Paquette. *SGD in the Large*. Conference on Learning Theory '21. Boulder.

- April 2021 ◦ E. Paquette. *The edge scaling limit of the Gaussian beta-ensemble characteristic polynomial*. Probability seminar. Jerusalem.
- January 2021 ◦ E. Paquette. *Secular coefficients and the holomorphic multiplicative chaos*. Analysis seminar. Tel Aviv University.
- December 2020 ◦ E. Paquette. *The edge scaling limit of the Gaussian beta-ensemble characteristic polynomial*. Séminaire MEGA (Matrices et graphes aléatoires). Institut Henri Poincaré.
- December 2020 ◦ E. Paquette. *The edge scaling limit of the Gaussian beta-ensemble characteristic polynomial*. University of Michigan seminar on integrable systems and random matrices.
- October 2020 ◦ E. Paquette. *Random perturbations of non-normal matrices*. **Queen's university colloquium**.
- October 2020 ◦ E. Paquette. *The threshold for simple-connectedness in hypercube percolation*. AMS "Southeastern" special session on random discrete structures.
- August 2020 ◦ P. Maillard and E. Paquette. *On a class of interval fragmentations with interaction between the fragments*. One world symposium 2020. Special session on branching and coalescing structures.
- March 2020 ◦ E. Paquette. *Random matrices and the Gaussian multiplicative chaos on the line*. UCSD probability seminar. San Diego.
- January 2020 ◦ E. Paquette. *Random matrices and the Gaussian multiplicative chaos on the line*. AMS Joint Mathematics meeting. Special session on random matrices. Denver
- October 2019 ◦ E. Paquette. *Random matrices and the Gaussian multiplicative chaos on the line*. CRM probability seminar. Montreal
- July 2019 ◦ E. Paquette. *The characteristic polynomial of random matrices and the Gaussian multiplicative chaos*. Random geometries and multifractality. International Institute of Physics. Natal, Brazil
- February 2019 ◦ E. Paquette. *The Gaussian analytic function is either bounded or covers the plane*. Weizmann Institute Probability seminar.
- February 2019 ◦ E. Paquette. *The characteristic polynomial of random matrices and the Gaussian multiplicative chaos*. McGill University.
- January 2019 ◦ E. Paquette. *Random matrix point processes via stochastic processes*, UC Irvine Probability Seminar,
- November 2018 ◦ E. Paquette. *Distributional approximation of the characteristic polynomial of a Gaussian beta-ensemble*, Cincinnati Symposium on Probability Theory, University of Cincinnati
- October 2018 ◦ E. Paquette. *Distributional approximation of the characteristic polynomial of a Gaussian beta-ensemble*, Courant Probability Seminar, NYU
- October 2018 ◦ E. Paquette. *Distributional approximation of the characteristic polynomial of a Gaussian beta-ensemble*, Wisconsin Probability Seminar, Madison
- September 2018 ◦ E. Paquette. *Random matrix point processes via stochastic processes*, AMS Sectional Meeting, Newark Delaware
- June 2018 ◦ E. Paquette. *Distributional approximation of the characteristic polynomial of a Gaussian beta-ensemble*, Gaussian Fields in Random Matrix Theory, Institute Mittag-Leffler

- May 2018    [○](#) E. Paquette. *Random matrix point processes via stochastic processes*, Weizmann
- May 2018    [○](#) E. Paquette. *Random matrix point processes via stochastic processes* , Southeastern Probability Conference, Duke
- April 2018    [○](#) E. Paquette. *Algebraic questions about combinatorial random matrices* , AMS sectional meeting, Nashville
- January 2018    [○](#) E. Paquette. *Perturbations of non-normal matrices* , AMS sectional meeting, UC Riverside
- November 2017    [○](#) E. Paquette. *Choices and intervals*, Georgia Tech Probability seminar
- October 2017    [○](#) E. Paquette. *The law of large numbers for the maximum of the log-potential of random matrices*, Northwestern University Probability seminar
- July 2017    [○](#) E. Paquette. *Tridiagonal models of  $\beta$ -Dyson Brownian motion*, PCMI Summer Session 2017: Random matrices
- March 2017    [○](#) E. Paquette. *Perturbations of non-normal matrices*, NYU Courant Probability seminar
- February 2017    [○](#) E. Paquette. *The law of fractional logarithm of the GUE minor process*, Michigan Probability seminar
- February 2017    [○](#) E. Paquette. *Probability and spectra*, Rabaden Lab, Columbia University
- January 2017    [○](#) E. Paquette. *Random perturbations of non-normal matrices*, UC Irvine Probability Seminar